

Observed metocean

ERA5/Ocean + NDBC
R-vine dependence

Physics response

5,000 low fidelity
400 OpenFAST cases

Reliability model

Response surrogate
operational system Pf

QAE evidence

Estimator studies
functional oracle

Optimization

ABC / DE controls
frozen candidates

Independent temporal validation (2020-2024)

600 s, ModCoupling=3 spot validation
4 designs × 3 independent wave seeds